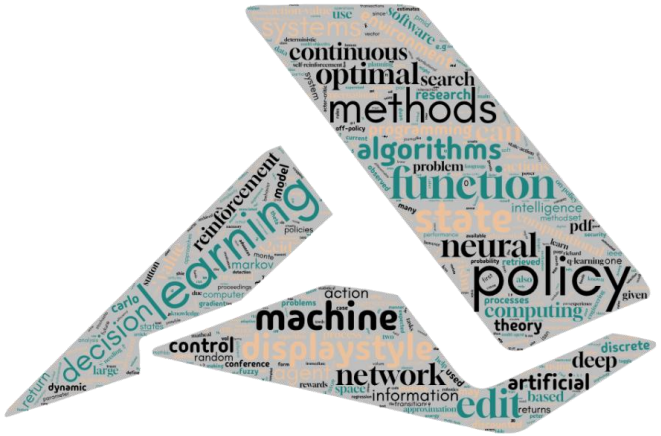
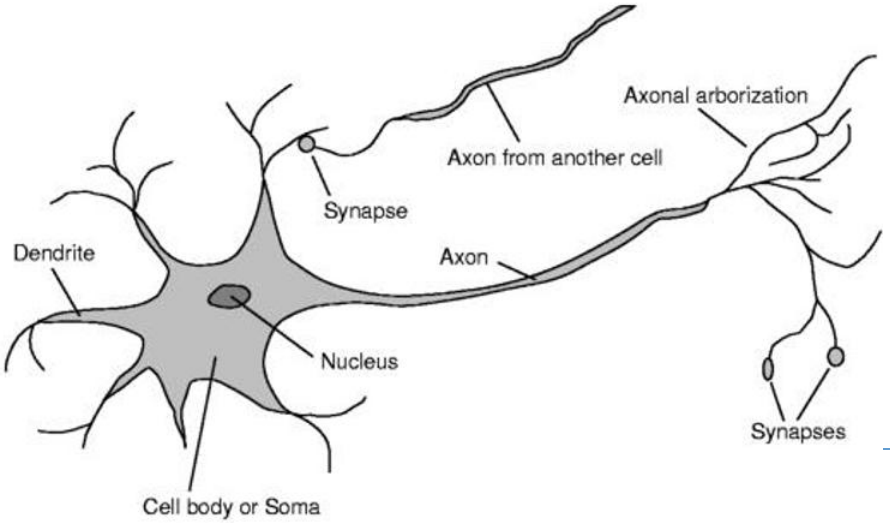


Reinforcement Learning Algorithms and Applications

Chapter 6. Neural Networks

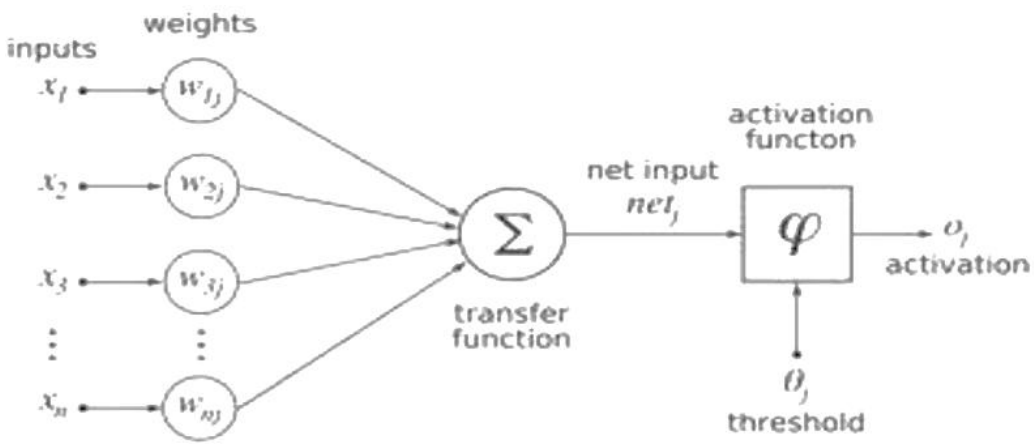
Artificial Neural Networks

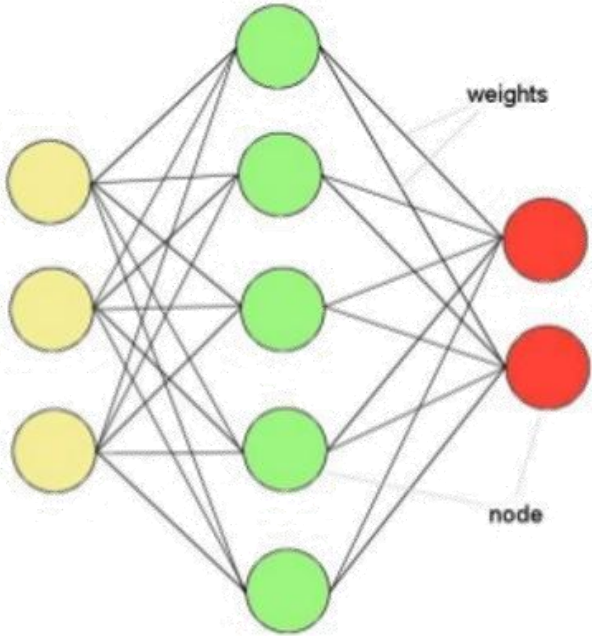
- What are they used for?
  - Classification
    - Pattern recognition, feature extraction, image matching
  - Noise Reduction
    - Recognize patterns in the inputs and produce noiseless outputs
  - Prediction
    - Extrapolation based on historical data



Artificial Neural Networks

- ANNs incorporate the two fundamental components of biological neural networks:
  - Neurones (nodes)
  - Synapses (weights)
- Neuron consists of three basic components.
  - Weights
  - Thresholds
  - Activation function

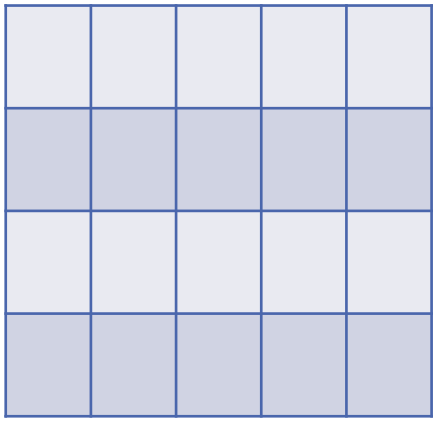




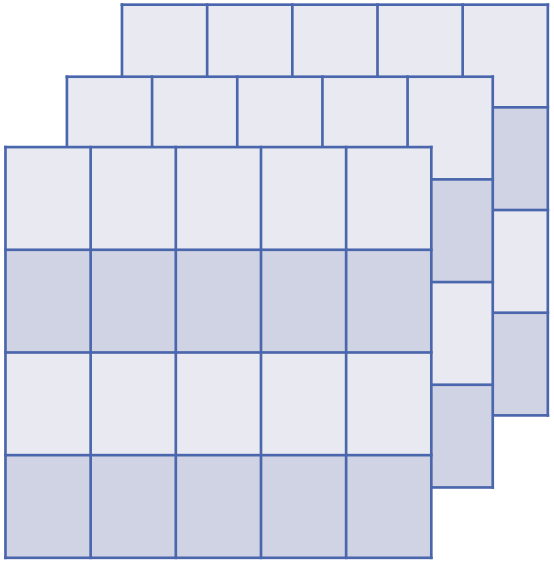
Tensors

- In TensorFlow or PyTorch, there are a lot of data encoded with tensors such as inputs and outputs of a model, as well as the model's parameters.
- Tensors can be calculated on GPUs and share the same underlying memory with NumPy arrays.
- What are tensors?

1-D Tensor
e.g. Audio



2-D Tensor
e.g. Grayscale Image



3-D Tensor
e.g. RGB Images

Reference

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